

MICROFLUIDICS IN DIAGNOSING ALZHEIMER'S

Group 4: Hot Jane Doe

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BACKGROUND IMAGE CREDIT: LABORATORY / ALAMY.

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1) Background

Alzheimer's disease is a progressive neurodegenerative disorder that gradually destroys memory, cognition, and independent function, leading to severe disability. It is the most common cause of dementia, typically developing in older adults and gradually worsening over time. Although it's proven that more women live with Alzheimer's disease in comparison to men (Beam et al., 2018), Alzheimer's is still the sixth leading cause of death, higher than other diseases whose death rates have declined due to improvements in detection and treatment. (*Alzheimer Disease*, 2025)

The number of people living with Alzheimer's or dementia nearly doubles every 20 years; this projects a total of 78 million people affected by Alzheimer's or dementia by 2030. (*Dementia statistics*, 2024) Early symptoms often include forgetfulness, confusion, and difficulty completing familiar tasks. Common warning signs include repeating questions, misplacing items, and difficulty recognizing familiar people. Unlike other diseases, the death rate of Alzheimer's disease has been steadily increasing each year. For example, from 2000 to 2001, reported deaths from the disease increased by more than 140%. ("2024 Alzheimer's disease facts and figures", 2024)

When designing a new tool for Alzheimer's diagnosis, it is important to prioritize features that increase patient comfort and remove elements that add unnecessary risk to the patient. Our design should have minimally invasive sampling, a low required sample volume, and a simple operation to minimize pain. With accessibility prioritized, the use of bulky or high-cost materials should be avoided. It should have portable and

user-friendly materials that can be used in different settings. The system should also only include the detection mechanisms necessary and leave out complex components that raise costs and slow down the overall process.

An effective method to help those with AD is to diagnose them early. To do this, pathophysiology, or bodily changes, must be understood. The two dominant biological hallmarks of Alzheimer's are senile plaques (SPs) and neurofibrillary tangles (NFTs). A β -42, a key component of SPs, is linked to mutations in the amyloid precursor protein, which produces excessive amyloid protein. There is naturally occurring A β -40 in the brain, which has a delicate ratio with A β -42. Compared to normal pathophysiology, the ratio of A β -42 to A β -40 is lower, indicating a sign of Alzheimer's. This causes neurotoxicity, leading to neurocognitive decline and dementia symptoms. The key component of NFTs is hyperphosphorylated tau proteins, and abnormal tau phosphorylation disrupts intracellular transport. When they spread to the neocortex, it worsens neurodegeneration. (Wu & Fuh, 2025)

These proteins (A β -40, A β -42, and p-tau), also known as biomarkers, are present in patients with Alzheimer's, allowing their detection to aid diagnosis of Alzheimer's. The two main sources of these biomarkers are through the blood and cerebrospinal fluid (CSF). Using CSF is advantageous because it has direct contact with the brain and contains central nervous system-specific proteins. (Hunter et al., 2025) Lumbar punctures are a common method of reaching CSF from the spinal canal, and the extracted fluid is analyzed for its biomarker ratios. (Hampel et al., 2022) Magnetic resonance imaging (MRI), computerized tomography (CT), and positron emission tomography (PET) are also commonly used to measure brain volume and

molecular activity, which provides another avenue for patient diagnosis. This method distinguishes the degree of degeneration from dementia, allowing doctors to discern patients in early or late stages of mild cognitive impairment. (Liu et al., 2022)

The current methods used for diagnosing Alzheimer's are not only lengthy but also costly. CT scans have a hefty direct cost of around \$1,942 to \$2,699, while the charges for MRI scans are even higher, ranging all the way from \$3,103 to \$4,487. (Kurniadi & Davis, 2022) The steep upfront cost of these scans limits access to the tests necessary to obtain a diagnosis. According to a study, observing the prices of neuropsychological evaluation versus neuroimaging, neurological testing appears to be comparable to neuroimaging. (Kurniadi & Davis, 2022) Looking beyond the initial costs of obtaining an Alzheimer's diagnosis, there's not much improvement in the standard of care of Alzheimer's patients. In fact, cost estimates of Alzheimer's underestimate the toll this disease takes on a patient and their caregivers by up to \$65,565 for mild Alzheimer's. (Fox et al., 2025) The cost for patients, beyond treatment, includes the cost of hospitalizations, medications, and home modification costs, which rise as the disease gets increasingly severe. (Tay et al., 2024)

Most microfluidic devices for Alzheimer's diagnosis are not injected into the body. However, by implanting a microfluidic device into the spinal cord, our proposed device can identify Alzheimer's as early as possible, allowing clinicians to intervene before irreversible neuropathological changes take hold. The motivation for this project is to design a microfluidic system capable of detecting key Alzheimer's biomarkers with high accuracy while minimizing patient risk, discomfort, and financial burden. This system

operates using minimal sample volumes and portable sensor technology, allowing clinicians to identify early changes long before structural brain degeneration is evident through imaging. This project aims to improve current diagnostic tools by offering a low-cost, minimally invasive option for early Alzheimer's detection.

2) Design Criteria and Constraints

Our proposed microfluidic device presents itself as a small, glass-based chip that is designed to detect biomarkers that are used to identify early Alzheimer's. Our device functions as an organ-on-chip containing microchannels with enzyme-coated sensing regions where small volumes of patient samples flow through the device. Proteins associated with Alzheimer's, such as A β -40, A β -42, and p-tau, interact with their corresponding immobilized enzymes. The chemical changes to the enzyme are transduced by electrochemical sensors to reflect biomarker concentration. These signals are amplified and transmitted as a radio frequency and received by a device such as a smartphone or smartwatch that will present the user with the results, allowing clinicians to receive real-time data. By combining biocompatible materials, precise biomarker detection, and wireless signal output, our design provides an accurate, cost-efficient, and minimally invasive device for Alzheimer's diagnosis.

A constraint related to the widespread manufacturing of microfluidic systems is the lack of a variety of resources used to create them and the high expense of certain materials. Thus, a large proponent of creating a socioeconomically favorable device is using items that are efficient, have multiple processes, and are generally inexpensive in comparison to their efficiency and significance to the product overall. (Shambhulinga Aralekallu, 2023) The device must be low-cost, durable, and long-lasting so that it can be widely distributed and used over time without losing functionality. The materials and production methods should allow for efficient manufacturing while maintaining quality, reproducibility, and stability during storage. (Moon, 2023) The effectiveness of the

product and safety towards the patient are qualities that trump any other aspects listed above, such as the cost. Any potential harm to the patient is avoided or minimized; thus, materials that may not be as financially favorable are still the most beneficial overall.

Additionally, the device must also meet certain functional requirements to effectively diagnose Alzheimer's. It must be highly sensitive to Alzheimer's biomarkers, A β -40, A β -42, and phosphorylated tau. (Sharma et al., 2023) Detecting minuscule amounts of these proteins is crucial for giving a correct diagnosis, and it reduces patient discomfort through efficiency. This sensitivity also helps separate Alzheimer's from other neurological disorders, ensuring that the results are precise and clinically accurate. Additionally, the device should focus on noninvasive operation, especially since it deals with biomarkers linked to the brain. (Wang et al., 2025) Reducing invasiveness lowers risk while making the diagnosis more comfortable for the patient. The efficient detection methods support this goal while still providing the detailed molecular information necessary to identify Alzheimer's early, even before visible brain atrophy happens.

One of the most important design requirements is biocompatibility, to help the system meet both medical and practical needs while remaining reliable for long-term use. Because the device encounters bodily fluids, it can't trigger immune responses or irritate tissues. Due to this, the materials and any coatings used need to stay stable over time and resist breaking down in the body's environment. (Tao Tang, 2021) Chemical stability is important so that fluid flow and biomarker detection can remain accurate. The device also has to handle all ranges of body temperatures and exposure to biological substances without swelling, softening, or losing strength within the body. Focusing on biocompatibility makes sure the device is safe, reliable, and effective

over time. The device also has to withstand mechanical stress from everyday body movements or accidental pressure. Strong, durable construction is essential both for keeping the device working properly and for protecting the patient.

When designing the microfluidic system, the chip's material is a major factor in its fabrication. There are a few materials that are typically chosen for the base of a microfluidic system, such as silicon, glass, polymers, and paper. These are popular choices due to their cost-effectiveness, malleability, high surface area, and a suitable pore size for gas permeability. (Shambhulinga Aralekallu, 2023) Despite the benefits of other materials, silicon and glass have more advantages; for example, some microfluidic systems need to withstand high temperatures, which polymers or papers can't. On the other hand, the stability of glass continues even up to 1400 degrees Celsius. Furthermore, the electronic components of the system are suitable for glass but are expensive and energy-consuming to implant into polymers and glass. (Shambhulinga Aralekallu, 2023)

A large proponent of creating a socioeconomically favorable device is to use items that can efficiently perform multiple processes and are generally inexpensive in relation to their efficiency and significance to the design and product overall. Using glass as the shell of the design is more advantageous in comparison to other materials, namely due to the incomparable transparency, small channel size, and compatibility with a vast number of organic materials. (Shambhulinga Aralekallu, 2023) From a socioeconomic standpoint, considering the longevity, efficiency, and advantageous nature of glass makes it have the highest benefit in comparison to other materials, and thus a greater

benefit long-term. For instance, while paper seems advantageous because it is one of the lowest-cost materials, its benefits in transparency and compatibility are lackluster compared to those qualities in glass. (Kidanemariam & Cho, 2025)

Generally, microfluidic systems are made of one of four different types of glass-based materials: borosilicate, quartz/fused silica, soda lime glass, or alkali-free glass. Despite each glass having its own merits, a major factor in choosing a glass body is biocompatibility because the microfluidic system is implanted in the patient's body. Of the four choices, quartz/fused silica has the highest biocompatibility, reported as an effective packaging to enclose the electronic and mechanical aspects of the device. (Joung, 2013) Other glasses are effective as microfluidic systems externally; however, most don't function reliably as a long-term implant. This is evidenced by the fact that soda lime and borosilicate glass can flake over long-term use in aqueous solutions, creating a risk to its user the longer it stays in the body. (Jung Eun Lee, 2023) Under the proper conditions, a quartz/fused silica chip can last decades unless physically damaged and tampered with, and the conditions it can withstand before sustaining damage are great. The material is very malleable, as well as adaptable to the required thinness of the final device. (Shambhulinga Aralekallu, 2023) Out of all glass materials listed, quartz/fused silica best satisfies the major requirements, those being transparency, temperature resistance, and flexibility.

The device must meet specific parameters to be compatible with the body, especially considering that it must be implanted into the body, so specificity in size takes great emphasis. The microfluidic system will have channel sizes of ~ 100 - $300\ \mu\text{m}$, ensuring a smooth liquid flow while keeping the design compact. (Tao Tang, 2021)

There are two main methods for device fabrication, including 3D printing and laser micromachining channels in the glass. These recent methods have improved their accessibility and efficacy, resulting in greater socioeconomic benefits. Glass-compatible 3D printing is a highly effective way to remain inexpensive, particularly due to the lack of outsourced materials and the efficiency of replication. 3D printers use sacrificial molds to create complex microstructures. (Jinhao Zheng, 2025) On the other hand, laser micromachining can be utilized for channel creation. It's a precise technique that utilizes laser beams to remove small amounts of material until microchannels are carved. The ultra-fast laser would be used because it has compatibility with quartz/fused silica, along with other glass types like porous glass and ultra-thin glass. (Tao Tang, 2021) Using an ultra-fast laser is likely an easier choice because glass-based printing is difficult to process. (Jinhao Zheng, 2025) The glass structure is created in 3D printing, whereas with a laser, the structure can be etched out.

Safety and performance are the top priorities, but the design also needs to be manufacturable at scale and affordable. Fabrication techniques should produce consistent, reproducible structures efficiently, and transparency can help with visual inspections during production or later monitoring. Balancing these factors helps create a device that isn't just effective for detecting Alzheimer's biomarkers but is also realistic to produce and use in a wider clinical setting, making early diagnosis more easily accessible to patients.

3) Design, Schematics, and Processes

Our microfluidic chip is a device that will be inserted through the spinal cord, and therefore, precision and size have to be highly specific. The general size of a microfluidic chip is relatively compact, ranging from 1 to 10 centimeters. However, due to the requirement of having to be as noninvasive as possible, using specifically a smaller value within the range, in this case, being 1 centimeter, would be best for the design. Additionally, quartz/fused silica is ideal for biocompatibility in the body to not trigger the white blood cells in the body, especially with its abundance in the cerebrospinal fluid of the spinal cord. (Jogia et al., 2021) Lastly, the microchannels in the glass will be etched out using an ultrafast laser, which is compatible with the material being used.

The microfluidic chip is designed to execute various tasks based on its specialization. In relation to the desired process of detecting biomarkers for Alzheimer's disease, the device will be used to detect proteins. Microfluidic devices are known to have strong and automated control of their microenvironment, and are therefore accurately able to detect many proteins. (Pattanayak et al., 2021) Alzheimer's disease is most closely connected to amyloid beta proteins and tau proteins, with amyloid beta proteins specifically being detected as a precursor to the disease. (*Alzheimer's disease*) The proteins that are typically seen as a precursor to the disease are A β -40 and A β -42 proteins. Thus, the microfluidic chip needs to prioritize detecting those proteins as part of the design. The systems used for detection within the microchip will be immobilized enzymes on electrochemical probes, which detect when the biomarkers bind to them. Normally, microfluidics systems for diagnosis use antibodies attached to an

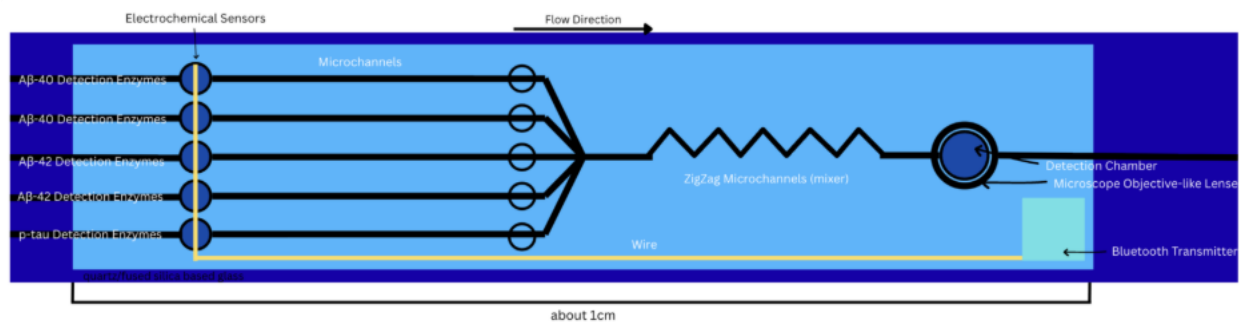
electrode because antibodies can easily attach to the biomarkers and produce an electrochemical signal. (Xinxiang Su, 2025) However, by using immobilized enzymes instead of antibodies, the biomarkers can be catalyzed and detached from the enzyme to be reused. Immobilized enzyme electrochemical biosensors have been used before to monitor food production processes, yet problems arise with the poor stability and limited lifetime of the enzyme. (Sun et al., 2023) Since the immobilized enzymes are contained in the chip and aren't exposed to as many factors as a food processing plant, the degradation of the enzymes would be more preventable. Testing should be used to determine which enzymes can still bind to the biomarkers while remaining cross-linked to the electrodes. Chemically bonding the enzyme to the electrode prevents detachment, but may alter the shape of the enzyme and affect enzymatic activity.

Additionally, p-tau proteins are also typically detected in association with Alzheimer's disease, and therefore should be programmed into the chip detection. This will indicate the progression of the disease, and thus, actions could be taken to mitigate the spread accordingly.

Another priority of this design was to use technology to send signals and data from the microchip directly to a device, such as a phone or computer. Thus, the design will include a strong radio frequency for Bluetooth connection. Wireless technologies have been used within implantable devices dating back to the 1950s, with remote and wireless treatment being successful in data collection for disease. (Nelson et al., 2020) This means the inclusion of Bluetooth for wireless connection is entirely feasible. Additionally, the method in which it will be added is through using a Nano FM

transmitter, which is typically sized at 3.5 mm and is therefore small enough to fit within the chip. The chemical change of the proteins, with the help of a transducer, would ideally send enough electric signal for the connection and data to be transmitted; however, if not, there will be signal amplification built in to mitigate any issues that may be caused if the signal is too weak.

Design used for implantation:



Channel Width: 50-200 μm , Channel Depth: 20-100 μm , Mixer: 6-12 zig-zag cycles, Detection Chamber Diameter: 0.5-2 mm, Sample Volume per Test: 1-10 μL

A whole host of tests need to be performed to ensure the device works before implanting it into a patient. Tests on the shape of the microchannels are needed to ensure the CSF can flow through uninterrupted, while the size of the microchannels needs to be tested to see if the CSF flows consistently.

Next, the enzymes bonded in the microchannels are to be tested. A known ratio of biomarkers should be injected into the microchannels to calculate if the ratio of the

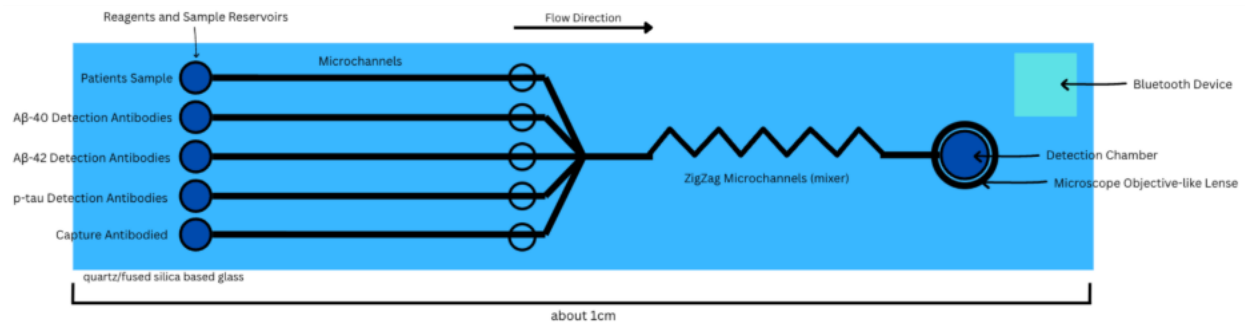
enzymes detected aligns. Along with that, we need to see if a signal is produced when the enzymes are in contact with biomarkers.

Lastly, the data given needs to be checked for accuracy and compared to a known result to guarantee the accuracy of the device.

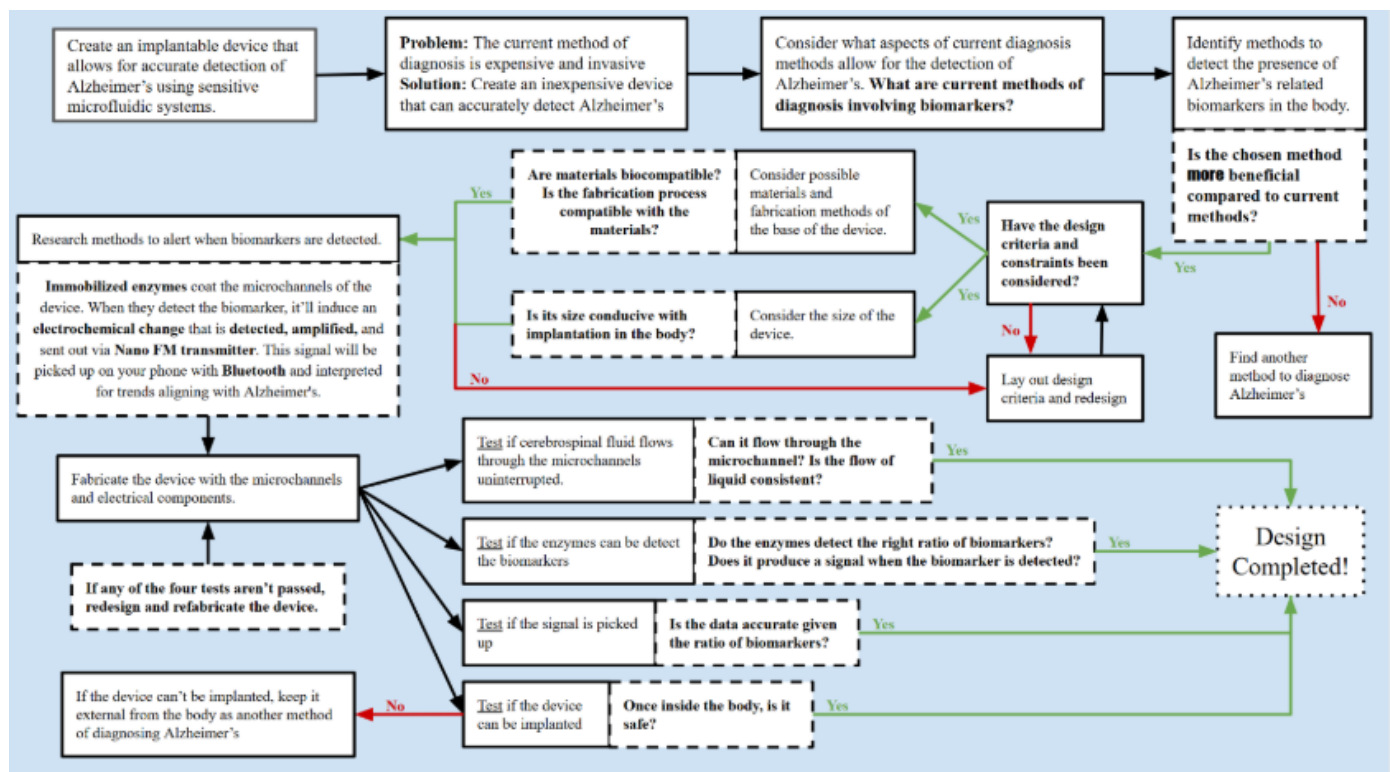
If our model isn't feasible enough to be implemented or if it fails during testing, the plan is to keep the device external. Arguably, the largest hurdle in our design is implanting the microfluidic device into a patient. If the device is too large or incompatible for implantation, the best option is to keep the device outside the body. The purpose of implanting the device was to ensure that the body's biomarkers could be continually tracked. This allows the patient to be diagnosed with Alzheimer's as early as possible because the patient is alerted as soon as the biomarker trend appears. This continuous tracking is what makes it better than a one-time lumbar puncture, MRI, or CT scan; it streamlines and quickens the process of diagnosis.

Even outside the body, without continuous tracking, the device still has functionality. Since the microfluidic device is extremely sensitive to the makeup of liquids, it is possible to use biomarkers in plasma from blood to diagnose AD. (Pais et al., 2023) Blood has more extraneous proteins compared to CSF, which is why plasma needs to be used instead of the blood itself. Nevertheless, plasma is considerably simpler, less invasive, and cheaper to obtain than CSF, which is why, even if not in the body, an external microfluidic device is still more beneficial than a lumbar puncture, CT scan, or MRI.

Design for external use:



Iterative design process:



4) Conclusion

Alzheimer's disease remains very prominent and difficult to diagnose early to this day. It entails current methods that heavily rely on expensive imaging, such as CT, PET, and MRI scans, and invasive cerebrospinal fluid testing. Our implantable design aims to approach these issues by fabricating a microfluidic device that minimizes invasiveness, cost, and creates an accessible platform allowing for the accurate continuous detection of biomarkers related to Alzheimer's pathology. By measuring the levels of A β -40, A β -42, and p-tau, the device can track electrochemical changes between these levels that signify early stages of Alzheimer's that occur before any cognitive symptoms appear.

This device is feasible due to advancements in microfabrication methods that are utilized throughout our design, such as laser micromachining. These procedures ensure biocompatibility as well as highly precise microchannels for the implantable system. This device would be considered clinically useful, as the continuous real-time data is received using only microliters of fluid, while also decreasing patient discomfort and reducing the need for expensive bioimaging.

However, despite the benefits that come with implantability, it also introduces some limitations. The immobilized enzymes that coat the microchannels may denature or lose activity over time. This could reduce the accuracy of biomarker detection. Thus, future improvements to this device may include enzyme stabilizing techniques to prevent decay or denaturing, a replaceable enzyme cartridge, or the creation of a synthetic material that could mimic enzymes that will last for a longer period. An alternative use of the device that would also work around this issue is to make the device external instead of implantable, eliminating this limitation completely.

Overall, this microfluidic device offers a promising approach to make early Alzheimer's detection more efficient, cost-friendly, less invasive, and accessible to society as a whole. Along with continuous refinement of enzyme stability and long-term sensor performance, this implantable device has a strong potential to improve clinical monitoring and patient outcomes.

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